

A photograph of a tree-lined street in New York City. The street is flanked by red brick buildings with multiple windows and fire escapes. Large, leafy green trees line both sides of the street, their branches arching over the road. Several cars are parked along the curb. A semi-transparent dark grey banner is overlaid across the middle of the image, containing the title text in white.

Exploring Tree Growth in NYC

Geocomp | Fall 2025

Research Questions

1. Which species of trees grew most in NYC?
2. Where did trees grow the most in NYC?

The Challenge: Linking Individual Trees Between Years



Method

If address (BBL#) and species are the same in 2005 and 2015, consider them the same tree.

Unless there are multiples in a year.

6. Create a TreeID across the years

```
[15]: # Add year column to each dataset
trees95['year'] = 1995
trees05['year'] = 2005
trees15['year'] = 2015
```

```
[16]: # Concatenate the datasets
trees_all = pd.concat([trees95, trees05, trees15], ignore_index=True)
```

---> Export trees_all concatenated dataset

```
[17]: #trees_all.to_file('data/cleaned_projected/trees_all.gpkg', Layer="trees", driver="GPKG")
```

```
[18]: # Give every unique BBL-spec combination a different code
trees_all["tree_id"]=(trees_all["bbl"].astype(str)+"_"+trees_all["spec_latin"]).factorize()[0]+1
```

```
[19]: # Separate back into individual years
trees95_id = trees_all[trees_all['year']==1995].copy()
trees05_id = trees_all[trees_all['year']==2005].copy()
trees15_id = trees_all[trees_all['year']==2015].copy()
```

```
[21]: trees05_id.shape
```

```
[21]: (592372, 13)
```

```
[22]: # Identify duplicates in each year
trees95_id["group_count"] = (
    trees95_id.groupby(["tree_id"])["tree_id"].transform("count")
)
trees05_id["group_count"] = (
    trees05_id.groupby(["tree_id"])["tree_id"].transform("count")
)
trees15_id["group_count"] = (
    trees15_id.groupby(["tree_id"])["tree_id"].transform("count")
)
```

```
[23]: # Remove duplicates
trees95_id = trees95_id[trees95_id["group_count"]==1].copy()
trees05_id = trees05_id[trees05_id["group_count"]==1].copy()
trees15_id = trees15_id[trees15_id["group_count"]==1].copy()
```

```
[25]: trees05_id.shape
```

```
[25]: (282673, 14)
```

```
[ ]: # Remove the group count column (drop column?)
trees95_id = trees95_id.drop(columns=["group_count"])
```

The Resulting Dataset

2005-2015 Trees with Linking TreeID

```
[11]: trees0515_poschange.shape
```

```
[11]: (101837, 23)
```

```
[12]: trees0515_poschange.columns
```

```
[12]: Index(['spc_common05', 'spc_latin05', 'tree_dbh05', 'year05', 'tree_id',  
       'spc_common15', 'spc_latin15', 'tree_dbh15', 'year15', 'geometry15',  
       'index_right', 'BoroCode', 'BoroName', 'CountyFIPS', 'NTA2020',  
       'NTAName', 'NTAAbbrev', 'NTAType', 'CDTA2020', 'CDTAName', 'Shape_Leng',  
       'Shape_Area', 'dbh_change'],  
      dtype='object')
```

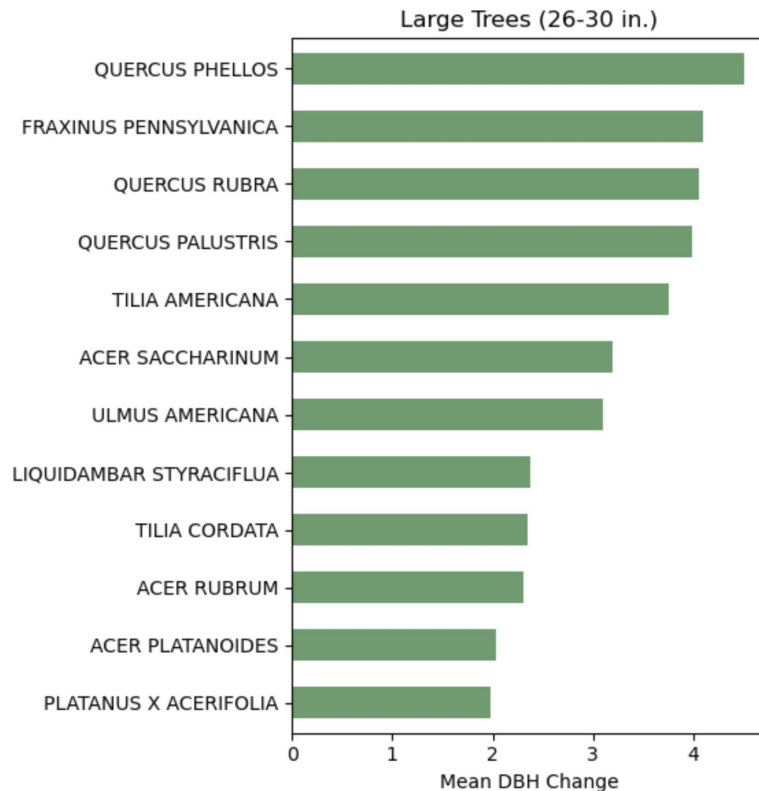
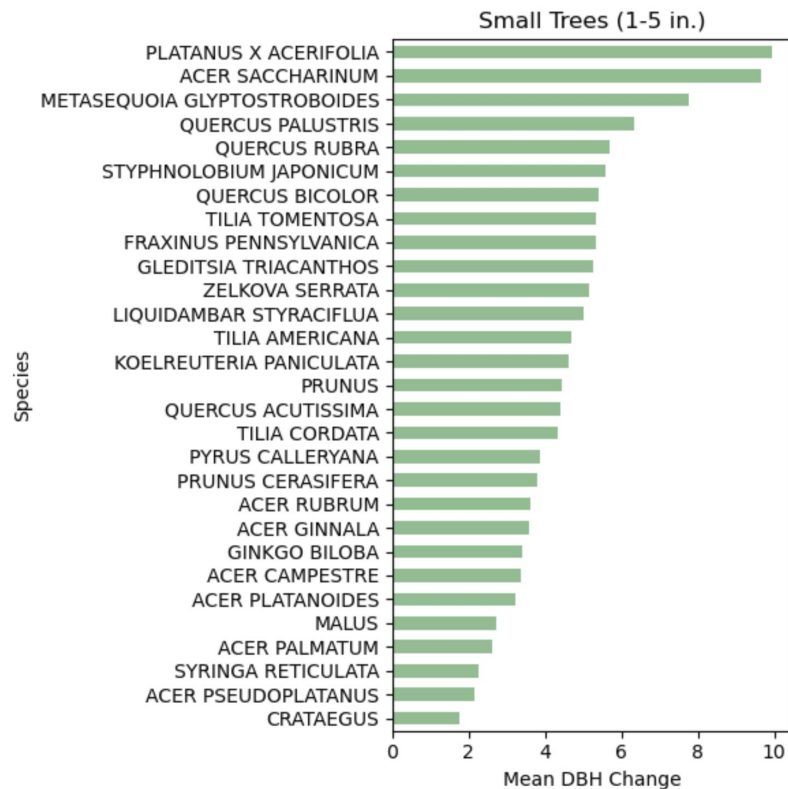
```
[43]: trees0515_poschange.sample(20)
```

```
[43]:
```

	spc_common05	spc_latin05	tree_dbh05	year05	tree_id	spc_common15	spc_latin15	tree_dbh15	year15	geometry15	...	CountyFIPS	NTA2020	NTAName
266141	LINDEN, LITTLE LEAF	TILIA CORDATA	4.0	2005.0	403977	LITTLELEAF LINDEN	TILIA CORDATA	17.0	2015.0	POINT (1019934.876 214727.139)	...	081	QN0303	North Corona
11039	LONDON PLANETREE	PLATANUS X ACERIFOLIA	15.0	2005.0	32439	LONDON PLANETREE	PLATANUS X ACERIFOLIA	18.0	2015.0	POINT (1020443.416 266483.47)	...	005	BX1203	Wakefield- Woodlawn
302818	PEAR, CALLERY	PYRUS CALLERYANA	4.0	2005.0	450215	CALLERY PEAR	PYRUS CALLERYANA	7.0	2015.0	POINT (940821.838 165299.125)	...	085	SI0107	Mariner's Harbor- Arlington- Graniteville
94961	HONEYLOCUST	GLEDITSIA TRIACANTHOS	7.0	2005.0	199127	HONEYLOCUST	GLEDITSIA TRIACANTHOS	10.0	2015.0	POINT (1014257.627 -----)	...	005	BX0202	Longwood

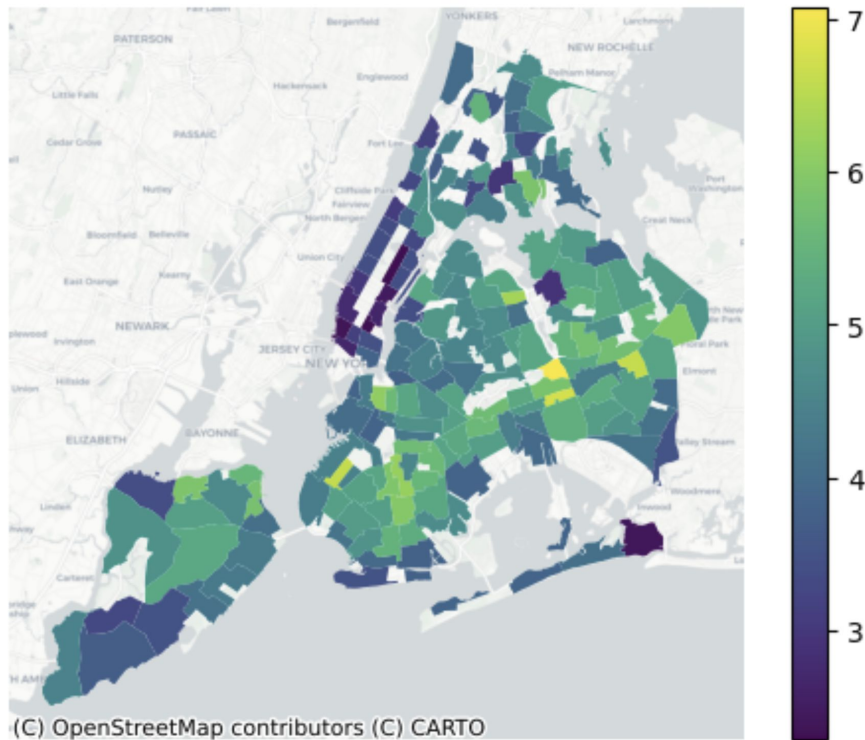
Results

1. Which species of trees grew most in NYC '05-'15?

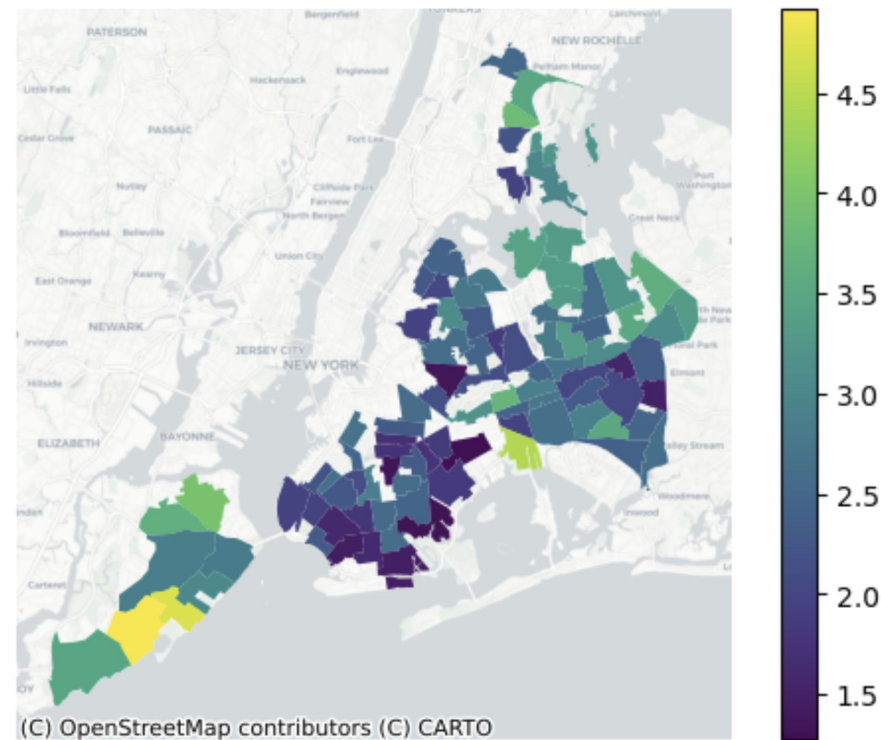


2. Where did trees grow the most in NYC?

Small Trees (1-5 in.)



Large Trees (26-30 in.)



Next Steps